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# EduOS Cameroon — Functional Requirements Specification

## Module 1: National School Registry (NSR)

|             |                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document ID | EDUOS-FRS-NSR-001                                                                                                                                      |
| Version     | 1.0 (Buildable Baseline)                                                                                                                               |
| Status      | Draft for Ministry Review                                                                                                                              |
| Supersedes  | Chapter 17 narrative specification (Volume II)                                                                                                         |
| Conventions | Identical to EDUOS-FRS-NTR-001: RFC 2119 keywords; every SHALL has an acceptance criterion; RFC 7807 errors; cursor pagination; OAuth2 via central IAM |

The NSR is the **foundational registry** of the platform: every transaction in every other module references a school. It covers **both ministries** (MINEDUB primary/nursery, MINESEC general + technical secondary) and both public and private schools, in a single ministry-scoped register — this is the concrete implementation of the shared-registry charter required by Risk R1.

### 1. Scope

**In scope:** school registration and lifecycle (open, merge, split, close); the National School Identifier (NSID); administrative hierarchy (region → division → sub-division → school); geolocation and accessibility classification; infrastructure and capacity profiles; enrolment summary data (per class level, per academic year); controlled update workflows with division-level validation; eligibility filtering for programmes; the public school directory; APIs consumed by every other module; carte scolaire data import.

**Out of scope:** individual student records (National Student Registry); staff HR records (Teacher Management); textbook stock (NTR/NWIDMS — the NSR stores no inventory).

### 2. NSID — National School Identifier (normative)

Format: `CM-SCH-{REG}-{DIV:2}{SUB:2}-{MIN}{TYPE}-{SEQ:5}` e.g. `CM-SCH-NW-0703-SG-00412`

| Segment | Values                                                                                                                                                                                          |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REG     | 2-letter region code (AD, CE, EN, ES, LT, NO, NW, OU, SU, SW)                                                                                                                                   |
| DIV/SUB | division and sub-division numeric codes from the national gazetteer (58 divisions / 360 sub-divisions, per BDA §1)                                                                              |
| MIN     | <input type="checkbox"/> B MINEDUB / <input type="checkbox"/> S MINESEC                                                                                                                         |
| TYPE    | <input type="checkbox"/> N nursery, <input type="checkbox"/> P primary, <input type="checkbox"/> G general secondary, <input type="checkbox"/> T technical, <input type="checkbox"/> C combined |
| SEQ     | 5-digit national sequence, never reused                                                                                                                                                         |

- **FR-NSR-ID-01** NSIDs SHALL be system-generated, immutable, and never reused. Existing ministry codes (MINEDUB/MINESEC references) SHALL be stored as `legacy_codes[]` aliases, searchable, so old paperwork remains resolvable.
- **FR-NSR-ID-02** If a school's administrative location is re-gazetted (sub-division split), the NSID SHALL NOT change; the hierarchy link changes with an audited effective date.

### 3. Data Model (normative)

```

AdminUnit (region 1–N division 1–N subdivision) ← reference data, gazette-versioned
Subdivision 1–N School
School 1–N SchoolStatusEvent (append-only lifecycle)
School 1–N InfrastructureProfile (versioned snapshots)
School 1–N EnrolmentReturn (per academic year, per class level)
School 1–N ProgrammeParticipation
School N–N School (merge/split lineage links)

```

### 3.1 SCHOOL (CORE FIELDS)

| Field               | Type                                                 | Req                      | Notes                                                                                                                                   |
|---------------------|------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| nsid                | string(24) PK                                        | ✓                        | §2                                                                                                                                      |
| name_official       | string(300)                                          | ✓                        | as gazetted/authorized                                                                                                                  |
| name_common         | string(300)                                          | ○                        | search alias                                                                                                                            |
| ministry            | enum {MINEDUB, MINESEC}                              | ✓                        |                                                                                                                                         |
| school_type         | enum {NURSERY, PRIMARY, GEN_SEC, TECH_SEC, COMBINED} | ✓                        |                                                                                                                                         |
| ownership           | enum {PUBLIC, PRIVATE_LAY, PRIVATE_CONF, COMMUNITY}  | ✓                        |                                                                                                                                         |
| language_system     | enum {EN, FR, BI}                                    | ✓                        | subsystem                                                                                                                               |
| subdivision_id      | FK AdminUnit                                         | ✓                        | full hierarchy derivable                                                                                                                |
| status              | enum §4                                              | ✓                        |                                                                                                                                         |
| authorization_ref   | string(100)                                          | ✓<br>for<br>pri-<br>vate | ministerial authorization number                                                                                                        |
| gps_lat / gps_lon   | decimal(9,6)                                         | ✓ *                      | *required for PUBLIC; captured on first field visit for others;<br>gps_source enum {GPS_DEVICE, MAP_PICK, IMPORTED} + gps_verified bool |
| accessibility_class | enum {URBAN, RURAL_ROAD, RURAL_SEASONAL, REMOTE}     | ✓                        | drives logistics planning (NWIDMS)                                                                                                      |
| grid_power          | enum {GRID, SOLAR, NONE}                             | ✓                        | drives solar-kit allocation (BUD §3.3)                                                                                                  |

| Field                        | Type                    | Req | Notes                    |
|------------------------------|-------------------------|-----|--------------------------|
| connectivity                 | enum {NONE, 2G, 3G, 4G} | ✓   | drives sync expectations |
| head_teacher_name<br>/ phone | string                  | ✓   | operational contact      |
| boarding, single_sex         | bool/enum               | ○   |                          |

### 3.2 INFRASTRUCTUREPROFILE (VERSIONED SNAPSHOT, ONE PER ASSESSMENT)

profile\_id PK, nsid FK, assessed\_at, assessed\_by, classrooms\_total, classrooms\_usable, has\_library, has\_lab\_science, has\_lab\_ict, storage\_rooms, storage\_secure (bool – precondition for school textbook stock), water\_source enum, latrines, fence (bool), source enum {SELF\_DECLARED, INSPECTION\_VERIFIED}

- **FR-NSR-DM-01** Infrastructure data SHALL carry its provenance (source); dashboards SHALL visually distinguish self-declared from inspection-verified data (Risk R8: no bad data presented as fact).

### 3.3 ENROLMENTRETURN

return\_id PK, nsid FK, academic\_year, class\_level (P1...P6/F1...F5/LS/US/C1...C4), boys, girls, submitted\_by, submitted\_at, validation\_status enum {SUBMITTED, DIVISION\_VALIDATED, REJECTED}, validated\_by

- **FR-NSR-DM-02** Enrolment totals per school-year SHALL be derived only from validated returns; unvalidated figures appear flagged. The validated series feeds the demand-forecasting module and NTR coverage (FR-NTR-15).

## 4. School lifecycle state machine

PROPOSED → AUTHORIZED → OPERATIONAL ⇌ TEMPORARILY\_CLOSED → CLOSED plus OPERATIONAL → MERGED / SPLIT (terminal for the record, with lineage links to successor NSIDs).

- **FR-NSR-SM-01** Merge SHALL require: target NSID(s), an effective date, and disposition instructions for open references (students, stock) — the API returns the list of open references from other modules (via their reference-check endpoints) and blocks merge while blocking references exist.

- **FR-NSR-SM-02** Only `division.validate` role may move a school to OPERATIONAL; only ministry-level roles may CLOSE/MERGE. All transitions are SchoolStatusEvents with actor + reference document.

## 5. Functional Requirements

| ID        | Requirement (SHALL)                                                                                                                                                                                                    | Acceptance criterion                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-NSR-01 | Register a school with mandatory §3.1 fields; duplicate-candidate detection at creation (name similarity $\geq 0.85$ trigram + same subdivision, or GPS within 500 m) presents matches before commit                   | Creating “GBHS Bamenda II” in a subdivision where “Government Bilingual High School Bamenda 2” exists surfaces the candidate; user must explicitly confirm “not a duplicate” (recorded) |
| FR-NSR-02 | Controlled updates: field-level edit permissions (school users edit contacts; division edits hierarchy; ministry edits status); every change versioned with actor/timestamp/old→new                                    | Change history endpoint returns full field-level audit; school user attempting hierarchy edit gets 403                                                                                  |
| FR-NSR-03 | Enrolment return submission per academic year and class level, with division validation workflow and rejection reasons                                                                                                 | Return with boys+girls exceeding school capacity $\times 1.5$ flags a warning; division rejection requires a reason code                                                                |
| FR-NSR-04 | Advanced eligibility search: filter by any combination of region/division/subdivision, type, ownership, status, enrolment range, infrastructure flags, accessibility, power, connectivity; results exportable CSV/XLSX | Query “MINEDUB PUBLIC, REMOTE or RURAL_SEASONAL, grid_power=NONE” returns the solar-kit target list (BUD §3.3)                                                                          |
| FR-NSR-05 | Public read-only directory (web + JSON) of OPERATIONAL schools: name, NSID, type, ownership, subdivision, GPS (public schools)                                                                                         | Unauthenticated GET excludes contacts and non-operational schools                                                                                                                       |
| FR-NSR-06 | Reference API for all modules: resolve NSID → school summary; batch resolve $\leq 1,000$ NSIDs per call; include <code>status</code> so consuming modules can block transactions against CLOSED schools                | NTR dispatch to a CLOSED school is rejected with <code>SCHOOL_NOT_OPERATIONAL</code>                                                                                                    |
| FR-NSR-07 | Carte scolaire bulk import (CSV, Annex template): staged validation report (duplicates, missing mandatory fields, invalid admin codes, GPS outside Cameroon bounding box) before commit; idempotent re-import          | Importing 22,000 rows with 300 defects commits 21,700 and produces a row-level defect report; re-running the same file creates zero duplicates                                          |

| ID        | Requirement (SHALL)                                                                                                                                                   | Acceptance criterion                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| FR-NSR-08 | Map view: schools plotted with cluster rendering; layers by status/type/coverage indicators; works on division-office bandwidth (tile caching)                        | 18,000 schools render < 5 s on reference hardware; offline tile pack available for field devices |
| FR-NSR-09 | GPS capture/verification from the field app: capture with device GPS (accuracy $\leq 25$ m recorded), verify existing coordinates, flag discrepancy > 1 km for review | Field verification creating a >1 km discrepancy opens a review case, does not silently overwrite |
| FR-NSR-10 | Programme participation: attach/detach schools to programmes (e.g., pilot wave 1, solar-kit wave 2) with effective dates; drives rollout management                   | Rollout dashboard counts by programme match attach records                                       |

## 6. Roles & permissions (summary)

| Action                                            | School user | Sub-div officer | Division officer   | Ministry admin | Public            |
|---------------------------------------------------|-------------|-----------------|--------------------|----------------|-------------------|
| Edit own contacts/infrastructure self-declaration | ✓           |                 |                    |                |                   |
| Submit enrolment return                           | ✓           |                 |                    |                |                   |
| Validate returns, register schools, verify GPS    |             | ✓<br>(propose)  | ✓                  | ✓              |                   |
| Status transitions<br>(operational/close/merge)   |             |                 | ✓<br>(operational) | ✓ (all)        |                   |
| Directory read                                    | ✓           | ✓               | ✓                  | ✓              | ✓ (public fields) |

Row-level scoping identical to FR-NTR-18 (server-side, own school / own division).



## 7. API summary

Base `/api/v1/nsr`: `/schools` (GET/POST), `/schools/{nsid}` (GET/PATCH), `/schools/{nsid}:transition`, `/schools/{nsid}/enrolment` (GET/POST), `/schools/{nsid}/infrastructure` (GET/POST), `/schools:resolve` (batch), `/schools:search`, `/directory` (public), `/imports` (staged carte scolaire), `/admin-units` (gazetteer reference). Same idempotency, pagination, and versioning rules as EDUOS-FRS-NTR-001 §7.

## 8. Non-functional requirements

| ID         | Requirement                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-NSR-01 | ≥ 40,000 school records (both ministries, public+private) with full history; reference resolution p95 ≤ 200 ms (it is on the hot path of every module)               |
| NFR-NSR-02 | Registry reads MUST remain available during central maintenance windows (read replica); consuming modules cache school summaries with 24 h TTL for offline operation |
| NFR-NSR-03 | Field app functions offline for registration verification and enrolment returns (same sync engine, FRS-NTR §9)                                                       |
| NFR-NSR-04 | Bilingual UI; gazetteer names stored FR + EN where official variants exist                                                                                           |
| NFR-NSR-05 | Personal data limited to head-teacher contact; no learner data in this module                                                                                        |

## 9. Data seeding & the R8 campaign

- **FR-NSR-MIG-01** Initial load = MINEDUB + MINESEC carte scolaire extracts (BDA §6.4) through the FR-NSR-07 pipeline, followed by the funded division-level validation campaign (BUD §3.5): each division receives its imported list, confirms existence/status/GPS of every school, and signs off. A school is `gps_verified` and eligible for device delivery only after this sign-off.
- **KPI linkage:** OUT-P7 (M&E) — data-quality score ≥ 85% pilot regions by end of the campaign.

## 10. Acceptance

100% SHALL requirements pass ACs in witnessed UAT; carte scolaire import executed on real ministry extracts with defect report reviewed; duplicate-detection precision/recall measured on a division-sized labelled sample (precision  $\geq 80\%$ , recall  $\geq 90\%$  at the FR-NSR-01 thresholds); reference API load-tested at 500 resolves/s.

## Annex — Import template `schools.csv`

```
legacy_code, name_official, ministry, school_type, ownership, language_system,  
region_code, division_code, subdivision_code, gps_lat, gps_lon, head_teacher_name,  
head_teacher_phone, enrolment_total_last_year, classrooms_total, grid_power, connectiv-  
ity, authorization_ref
```